

Harshal Abdulla

Associate / graduate software engineer · backend-leaning full stack

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MSc Computer Science graduate who builds systems where being wrong is expensive. I built and maintain a commercial ordering system that takes real money in a restaurant every day, and wrote the property-based test suite that proves its pricing is correct. Looking for an associate or graduate software engineer role in Ireland. Available now.

SELECTED PROJECTS

Restaurant ordering and till system

Commercial, sole developer · 2026

Client system running daily in a restaurant in Co. Kildare

- Built four surfaces (customer kiosk, kitchen till, table service, admin) from one Expo codebase against one Postgres database, in seven weeks, as the only developer.
- Put pricing and entitlement rules in plpgsql behind *security definer* so all four surfaces get the same answer and no client can bypass the till. Around 70 Postgres functions behind row level security across 94 migrations.
- Removed the application server entirely: static hosting plus row level security, with the client talking to Postgres through PostgREST and RLS as the security boundary.
- Money is integer euro cents with no floating point in pricing. Order lines are immutable snapshots and corrections are soft deletes, so history is never rewritten.
- Wrote a property-based test suite from scratch with no test framework and no dependencies: 43 properties, 560,015 generated cases per run in about a second, seeded so any failure replays exactly.
- Measured an order-dependent pricing effect across 20,000 baskets, showed the owner the numbers, and locked the guarantee it rests on into a test that runs on every change.
- 154 commits. 15,109 lines of TypeScript. 251 orders and 1,474 dishes in August 2026.

Stack: TypeScript 5.9, React 19.2, React Native, Expo SDK 54, Zustand, PostgreSQL via Supabase, row level security, pg_cron ·
Demo: order-dependent-pricing.surge.sh

Fault-tolerant notification system

Personal · in progress

- Distributed delivery pipeline built so a message is never silently lost and never sent twice, even when the broker, database or consumer crashes mid-flight.
- Producer writes a PENDING row to Postgres *before* publishing to Kafka, so a crash between the two leaves a durable record that a recovery sweep picks up.
- Consumer reads row status before acting (SELECT before UPDATE), which makes it idempotent by design and absorbs Kafka's at-least-once duplicates as a no-op.
- Failure path designed but not yet built: exponential backoff with a dead-letter queue, and retry counters in Redis rather than process memory, because process memory resets on restart and parallel consumers would each keep a separate count.

Stack: Python, Apache Kafka, PostgreSQL, Redis, Docker Compose · github.com/Harshal-Abdulla/Kafka-Notification-System

Sketchpad Retro

Individual MSc project · 2025

- Browser rebuild of Ivan Sutherland's 1963 Sketchpad, with the constraint solver, geometry engine and undo/redo stack all written from scratch and the interface drawn in SVG.
- Constraints including parallelism, perpendicularity, equal length and fixed distance, plus snapping, zooming, symbols and instances, and state persisted via JSON and local storage.

Stack: Next.js 14, React, TypeScript, SVG

Twix

Group MSc project, team of five · December 2024

- Led a team of five building a Twitter-style microblog: search for a user, follow and unfollow them, post tweets up to 250 characters, read a feed of who you follow.
- My own share of the work was the interface and the testing. Based the layout on Twitter so all five of us had a single reference to build against, rather than designing it from scratch and arguing about it.

- Followers are notified through an observer-pattern implementation in the backend. The feed polls the REST API every three seconds.

Stack: Java 21, Spring Boot 3.4, PostgreSQL, Spring Data JPA, React, TypeScript, Vite, Axios, Bootstrap · github.com/Harshal-Abdulla/twix

SyncUp

Group MSc project · 2025

- Task tool with group collaboration: tasks are private, shared with a group, or public, each can be tagged and marked complete. Full CRUD over a REST API, with password hashing using bcryptjs.

Stack: Node.js, Express, MongoDB, React

EXPERIENCE

Frontend Engineer, Cofount Labs LLP

September 2023 – August 2024 · Remote

- Built and shipped client-facing web applications for agency clients.

EDUCATION

MSc Computer Science (Software Engineering)

2024 – 2025

Maynooth University, Ireland · First Class Honours

Software testing and quality assurance, requirements engineering and system design, advanced object-oriented programming and scalable systems, distributed systems, applied cryptography, deep learning foundations.

B.Tech Computer Science and Technology

2019 – 2023

Jyothi Engineering College, Kerala, India

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900)

March 2026

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, SQL, Java
Frontend	React, React Native, Next.js, Expo, Zustand, Tailwind CSS
Backend & data	Node.js, Express, Spring Boot, PostgreSQL, Supabase, MongoDB, Redis, REST
Infrastructure	Apache Kafka, Docker Compose, GitHub Actions, Git
Testing	Property-based testing, test design, row level security policy testing